# DATA CLEANING PROJECT

CODING(main.py):

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("data.csv")

# Show first 5 rows
print("First 5 rows:")
print(df.head())

# Check missing values
print("\nMissing values:")
print(df.isnull().sum())

# Fill missing values
df = df.ffill()

# Basic info
print("\nData Info:")
print(df.info())

# Statistics
print("\nSummary:")
print(df.describe())

# Average salary
print("\nAverage Salary:", df['Salary'].mean())

# Department wise salary
print("\nDepartment-wise Salary:")
print(df.groupby('Department')['Salary'].mean())

# Gender count
print("\nGender Count:")
print(df['Gender'].value_counts())

# Visualization
sns.histplot(df['Age'])
plt.title("Age Distribution")
plt.show()

df.groupby('Department')['Salary'].mean().plot(kind='bar')
plt.title("Avg Salary by Department")
plt.show()
```

OUTPUT(python shell):

```
First 5 rows:
  Name   Age  Gender Department   Salary  Experience
0    A  23.0    Male         IT  20000.0           1
1    B  24.0  Female         HR  25000.0           2
2    C  25.0    Male    Finance  30000.0           3
3    D   NaN  Female         IT  28000.0           2
4    E  27.0  Female         HR      NaN           4

Missing values:
Name              0
Age               2
Gender            0
Department        0
Salary            2
Experience        0
dtype: int64

Data Info:
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Name        15 non-null     str
 1   Age         15 non-null     float64
 2   Gender      15 non-null     str
 3   Department  15 non-null     str
 4   Salary      15 non-null     float64
 5   Experience  15 non-null     int64
dtypes: float64(2), int64(1), str(3)
memory usage: 852.0 bytes
None

Summary:
             Age        Salary  Experience
count  15.000000     15.000000   15.000000
mean   26.400000  30333.333333    3.533333
std     2.640346   6320.789583    1.922300
min    22.000000  20000.000000    1.000000
25%    24.500000  26500.000000    2.000000
50%    26.000000  30000.000000    3.000000
75%    28.500000  34000.000000    5.000000
max    31.000000  42000.000000    7.000000
```

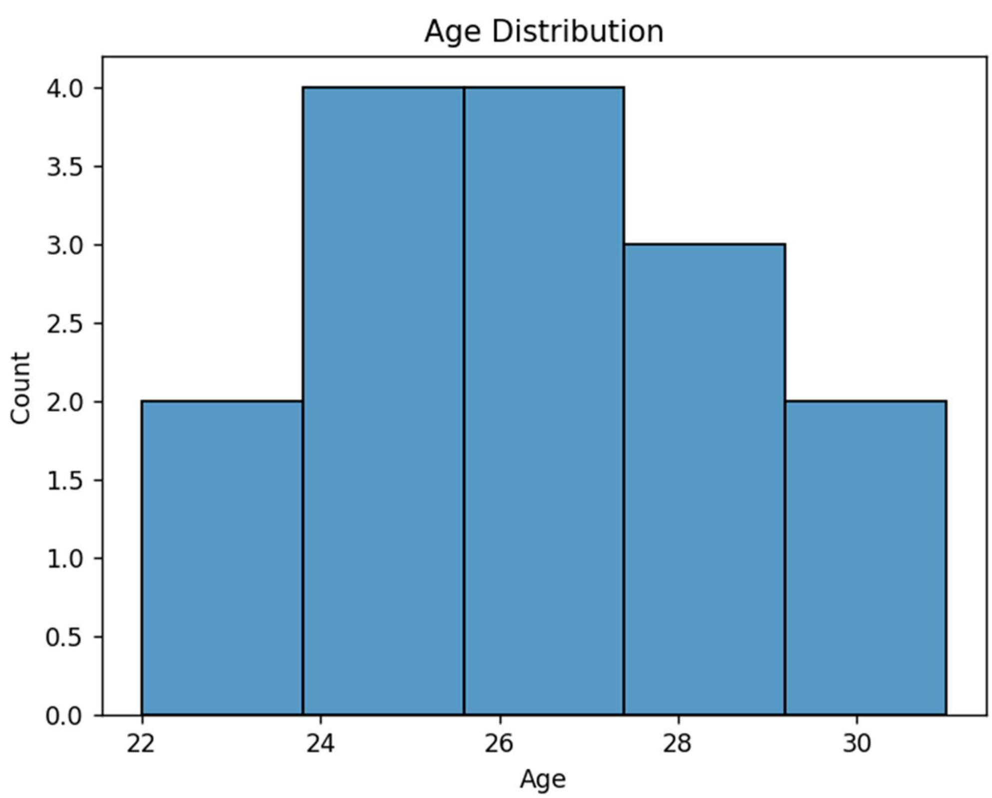
```
Average Salary: 30333.333333333332

Department-wise Salary:
Department
Finance    31800.0
HR         30800.0
IT         28400.0
Name: Salary, dtype: float64

Gender Count:
Gender
Male      8
Female    7
Name: count, dtype: int64
```

GRAPH(visualization):

1.Age Distribution



2.Average Salary by Department